CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm

(Clinical Pharmacy & Pharmacology and Toxicology)

University Theory Examination January 2018

PHCC004 ANALYTICAL TECHNIQUES

Date: 01.01.2018 , Monday Time: 10.00 a.m. to 01.00 p.m Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1	**A)**	Enlist the UV spectrophotometric methods for multicomponent analysis. Write a note on any one of them.	**05**
	B)	Enlist various inonizers used in Mass Spectroscopy. Write a note on MALDI.	**05**
Q.2	**A)**	What is NMR? Write a note on factors affecting chemical shift.	**05**
	B)	Write a note on sampling techniques used in IR spectroscopy.	**05**
		OR	
Q.2	**A)**	Discuss the advantages of UV spectrophotometric method over separation techniques. Write a note on solvents used in UV spectrophotometric methods.	**05**
	B)	Enlist factors affecting fluorescence spectroscopy. Discuss the instrumentation and applications of spectrofluorimeter**.**	**05**
Q.3		**Answer the following (Any three)**	**15**
	(A)	Discuss in brief cell imaging techniques.	
	(B)	Write a note on flow cytometry.	
	(C)	Write a note on fragmentation rules in mass spectroscopy.	
	(D)	Write a note on FT-NMR technique.	

SECTION-II

Q.4	**(A)**	What is chromatography? Discuss stationary phases used in GC.	**05**
	(B)	Discuss the principle of Ion exchange and Size exclusion chromatography.	**05**
Q.5	**(A)**	Explain: LC-MS, Flash chromatography	**05**
	(B)	Differentiate HPLC and HPTLC. Write applications of HPTLC.	**05**
		OR	
Q.5		What is Isoelectric focusing? Discuss in brief. Write applications of electrophoresis.	**10**
Q.6		**Answer the following (Any three)**	**15**
	(A)	Write note on RIA.	
	(B)	Write brief note on Immunofluorescent microscopy.	
	(C)	Discuss in situ gene expression technique.	
	(D)	Discuss capillary electrophoresis.	

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